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An improved data-driven method for the prediction of elastic properties in unconventional shales from SEM images

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A B S T R A C T

This paper demonstrates a quick look approach to estimate rock mechanical properties, such as Young's modulus, from SEM images of drill cuttings acquired from several unconventional plays across North and South America. Unlike traditional methods that involve extensive lab measurements or interpretive well logs, our approach leverages image-based analyses, significantly reducing the subjectivity and computational burden often encountered in previous strategies.

In our analysis, we processed SEM images from 14 plays to extract both textural and shape-based features. Textural attributes such as entropy, homogeneity, contrast, and energy provide insights into the disorder and mineralogical contrasts within the rock. There exist several shape-based features such as area, aspect ratio, circularity, solidity, extent, eccentricity, Euler number, and orientation which can describe the geometric properties of mineral constituents. We utilize both textural attributes and pore aspect ratios and integrate deep learning inputs to construct various predictive models.

Our models correlate these features with empirically measured Young's modulus values using non-parametric regression. This integrated approach has shown to provide a robust and generalizable model capable of estimating Young's modulus across a diverse set of geological formations with high reliability, even when tested against previously unseen images.

This study acknowledges that mineralogy and the juxtaposition of various minerals may be unable to fully account for the variations seen in the Young's modulus. Complex pore systems such as lenticular pores may lead to overestimation of elastic moduli. Therefore, the inclusion of both textural and shape attributes as proxies for mineralogy and their spatial arrangement addresses key controls in the mechanical behavior of rock samples, thereby enhancing the model's applicability across varied mineralogical and porosity conditions.

Our findings indicate that the combination of texture and shape analyses, coupled with machine learning techniques, can efficiently and accurately predict mechanical properties in tight rocks. This method represents a significant advancement over traditional approaches, providing a fast, non-subjective, and computationally efficient tool for preliminary rock mechanics analysis. This work underscores the potential of using SEM image analyses as a powerful tool for rapid screening and detailed rock mechanics studies, moving towards more streamlined and data-driven exploration and production strategies.

1. Introduction

Unconventional shales continue to be a topic of active research because of their complex microstructure that impacts both the transport and storage of fluids. Imaging techniques such as scanning electron microscopy (SEM) have allowed the visualization of microstructure and facilitated efforts to tie microstructure with petrophysical properties. The microstructure of shale consists of inorganic minerals such as quartz, clays, calcite, and organic matter that compose shale, the size and shape of these materials, how they are arranged together, and the porosity associated with them. This microstructure is influenced by depositional environment and can exhibit a high degree of heterogeneity that varies from play to play (Curtis et al., 2012) and within a play. Additional complexities include the presence of organic-hosted, inorganic-hosted, or mix-hosted pores that complicate quantitative image interpretation. These challenges have spurred the development of machine learning-assisted workflows to assist in SEM image analysis (Knaup et al., 2022; Li et al., 2019; Cang et al., 2017).

While artificial neural networks (ANN) formed the backbone of several image analyses workflows up until the early 2000s, they have since been succeeded by convolutional neural networks (LeCun and Bengio, 1995). Among several of their advantages is that convolution can process images so that the outcomes are translation invariant. This has led to wide adoption of CNNs for image classification and segmentation over the past decade and are considered to be one of several deep learning approaches (Szegedy et al., 2014; Krizhevsky et al., 2017). These techniques are now being employed in various fields such as medical imaging informatics (Kiran Reddy Maryada et al., 2022; Mudduluru et al. 2023; Danala et al. 2022) and the oil and gas industry (Maniar et al., 2018; Parshall, 2018; Araya-Polo et al., 2020; Knaup

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et al., 2022).

Unfortunately, CNNs also require vast amounts of training data. This can be a significant constraint for most applications, such as those that rely on the use of high-resolution, high-quality SEM images (Knaup et al., 2022). Overcoming this constraint requires the generation of synthetic training data such as discussed in Li et al. (2019), Cang et al. (2017), Kamrava et al. (2019), Kiran Reddy Maryada et al. (2022) where new images were generated by learning from a limited high-resolution dataset for use in training a CNN. Nevertheless, these approaches are computationally expensive and can require prohibitively large amounts of time and effort to ensure the regenerated images are of acceptable quality. Even with a large database which has its own storage requirements, it can be a tedious task to train a CNN, requiring access to computational resources that are beyond the capabilities of most desktop computers.

Prior approaches (Madadi et al., 2009; Arad et al. 2010) that have extracted mechanical properties such as elastic moduli from images have shown promise but are computationally challenging to implement in practice. They rely on mineral and pore segmentation using grain partitioning techniques. The results are then used in a finite-element model to simulate response of the porous media to various stresses from which the elastic properties are computed. The key drawback of this approach is the assumption that the mechanical properties of the individual grains are well known allowing for a simulation of a composite system response. For multi-mineral complexes, the predictions can be associated with a high degree of uncertainty because individual mineral properties are often unknown.

In a previous study (Yang et al., 2023), we showed how textural attributes can be used to predict the Young's modulus for fine-grained rocks. We also demonstrated that, contrary to what is generally assumed, the use of mineralogy as well as textural attributes to predict the Young's modulus did not provide any additional benefit over using just the textural attributes alone. The conclusion from the study was that texture, captured from the SEM images, is sufficient to capture the mineralogy (through variations in grayscale levels), but more importantly, provides a gross estimation of the spatial arrangement and heterogeneity of the different minerals. Our understanding is that texture alone captures the mineralogical variations within the samples without the need for explicitly estimating/measuring mineralogy.

In this paper, we propose an enhanced approach that relies on extracting information related to image texture and shape-related attributes to estimate rock mechanical properties. This stems from the need to account for the varying impact of circular pores and lenticular or high-aspect ratio pore systems. The pore shape-related attributes are shown to be crucial to enhance the accuracy of our predictions. We investigated several different models. First, we use only image textural attributes; then we fuse image-based texture and pore-derived shape-based features; for a third model, we also investigate any additional contributions with the use of deep learning features from a classifier trained to identify the source (i.e. the play) of the image. With the use of deep learning features, we avoid the high dimensionality of the inputs by pairing with two dimensionality reduction techniques (PCA and random projections (Dasgupta, 2000)). Finally, we use a machine learning model to predict rock mechanical properties such as Young's modulus. While we use SEM images for our analyses, the approach can be generalized to larger-scale images such as those obtained from thin sections. The approach presented here allows us to process images from samples like drill cuttings for the rapid prediction of mechanical properties like the Young's modulus.

The paper is organized as follows: We first provide a brief description of the dataset including the source of the images and the target variable which is the Young's modulus. We then discuss texture attributes and texture analyses of images, specifically in the context of SEM images acquired from samples collocated with those used in the lab. We describe the shape-related attributes used in the paper, following which we present a brief discussion of the deep learning network used in the paper. Finally, we present our results and concluding remarks.

2. Description of the dataset

Table 1 shows the list of unconventional plays used in this study. We extracted the Young's modulus using nanoindentation and the mineralogy of the samples, measured separately using FTIR (Fourier Transform Infrared Spectroscopy). This information alone could be utilized to relate mechanical properties to mineralogy. However, our work does not rely on the explicit use of mineralogy but uses texture to capture the spatial variation in mineralogy as well as the predominant shape of the pore systems.

We then used SEM images collocated with the samples used for lab-based nanoindentation Young's modulus measurements and analyzed them for the Haralick textural attributes as well as pore aspect ratios. In the case of more coarse-grained samples, the measurements of elastic properties would be acquired from more conventional lab-based approaches, such as a hydraulic press, as opposed to nanoindentation experiments which are suitable only for fine-grained rocks.

The SEM image dataset for each formation was gathered systematically using FEI MAPS software on a FEI Helios Nanolab 650 DualBeam system at 2 kV accelerating voltage and 0.34 nA current. Individual images in the data sets collected were 1536 x 1024 pixels. However, random crops of these larger images were made to collect over 3000 224x224 pixels at 50 nm resolution.

3. Description of key concepts

Image-Based Textural Attributes: In image processing, texture refers to the spatial distribution and arrangement of image intensity levels.

Prior approaches for microstructural analyses and extraction of petrophysical properties such as porosity or proportion of different pore systems or even mineralogy have relied on exploiting the information embedded in grayscale intensity (Knaup et al. 2019). However, simple grayscale intensity does not lend very well to the prediction of mechanical properties. A simple explanation is provided through Fig. 1. The three images can be seen to be characterized by different textures, while on the right we observe the corresponding grayscale intensities. Although the spatial arrangement of grayscale values is different, the grayscale intensity histograms (as a proxy for mineralogy) are identical. It is quite likely, however, that the mechanical behavior of the three samples whose images are shown are different – simply because of the varying spatial arrangement of the grayscale levels.

Other petrophysical properties such as porosity or mineralogy estimation from images essentially rely on point counts of pixel grayscale values. In back-scattered electron (BSE) SEM images, these grayscale intensities contain a wealth of information related to mineralogy with higher atomic weight material being lighter and low atomic weight material (such as organics) appearing darker and consequently, can be used to estimate mineralogy. However, this approach cannot be generalized to the image-based estimation of mechanical properties which are governed by mineralogy and porosity (both of which can be obtained from grayscale analyses), and more importantly, the spatial arrangement of the grains and pores. It is not hard to imagine that the mechanical properties of the samples (from the Eagle Ford and Haynesville) shown in Fig. 2 are substantially different because of a difference not

Table 1
The list of 14 plays used in this study.

Alum	Kimmeridge
Barnett	La Luna
Collingwood	Marcellus
Duvernay	Niobrara
Eagle Ford Gas	Osage
Eagle Ford Oil	Utica
Haynesville	Vaca Muerta Gas

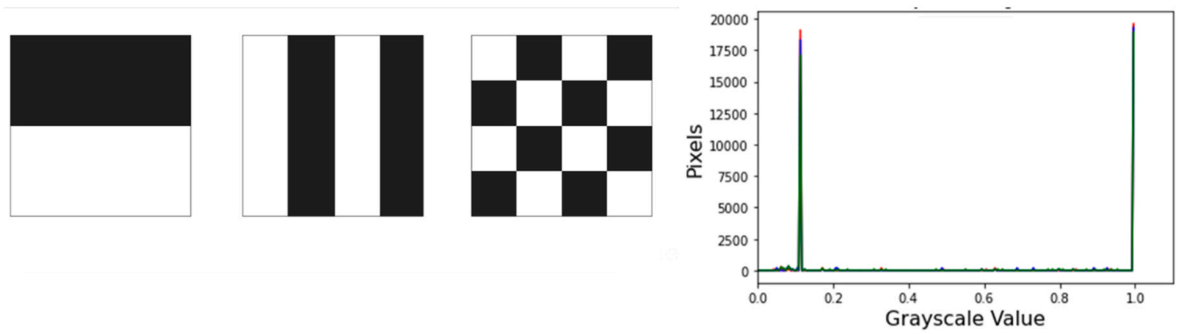

Fig. 1. The three images on the left are associated with different textures (or arrangement of grayscale values). The grayscale histogram on the right shows that all three images are similar with respect to the abundance of each grayscale value. Yet, it is not hard to imagine that the three images (from three different samples) would behave differently under any kind of stress. This underscores the need for including more spatial information for the prediction of elastic moduli from images.

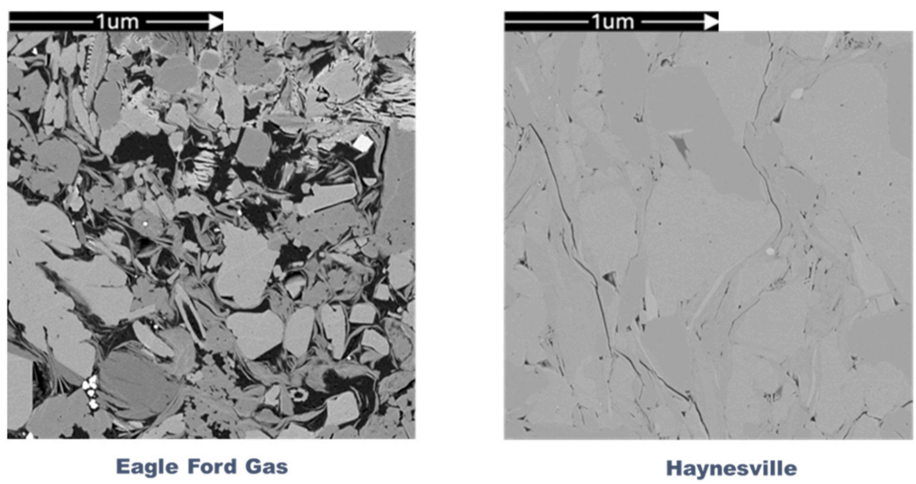

Fig. 2. SEM images of samples from the Eagle Ford on the left and the Haynesville on the right show differences in grayscale intensities and spatial arrangement. It is very highly likely that their mechanical properties will be different.

only in grayscale intensities but also because of the spatial arrangement of the grayscale values.

The example shown in Fig. 2 motivates a closer look at texture. For image analyses, the information related to texture (grayscale frequency and spatial arrangement) is embedded in a matrix known as a Gray Level Co-Occurrence Matrix (GLCM) (Haralick et al., 1973). The 2-dimensional GLCM provides information about the locations of pixels of similar grayscale values and how they are juxtaposed with other grayscale intensities.

A simple computation of the GLCM is shown below in Fig. 3. There are three grayscale images on the left, while the middle column shows the pixel grayscale intensity values corresponding to the image on the left. The matrix on the right is the GLCM matrix. For simplicity of illustration, pixels are said to be neighbors if they are next to each other in the horizontal direction. The example in Fig. 3 does not consider the GLCM computation in other directions for simplicity of illustration. When the image is uniformly black with 12 pixels of intensity 0 (image in the top row), the computed GLCM shows 12 pixels of intensity 0 adjacent to pixels of intensity 0. For the image in the middle row, there are 6 pixels of intensity 0 adjacent to pixels of intensity 0 and another 6 pixels of intensity 3 adjacent to pixels of intensity 3. For the image in the bottom row with a heterogeneous arrangement of pixel grayscale values, the GLCM matrix provides the corresponding spatial arrangement of grayscale values. We can now analyze the GLCM matrix to provide information on not just the grayscale intensities but also the spatial arrangement of grayscale intensities. For instance, in the bottom row, 1 pixel of intensity 3 is adjacent to a pixel of intensity 2.

In this manner, GLCM matrices can be constructed for grayscale SEM images of rock samples which would then provide the grayscale intensities and how the minerals are juxtaposed with each other. From these GLCM matrices, we can then extract textural attributes such as contrast, correlation, dissimilarity, entropy, energy, and homogeneity. The reader is referred to Haralick et al. (1973) paper for details on the computation of these values. In our workflow, to account for differences in acquisition settings for the images, which were collected over a period of 11 years using three different SEMs, we standardized all images to the same grayscale range. No other data augmentation was necessary.

Fig. 4 plots the measured Young's modulus against the various textural attributes used in this study. Given the lack of any obvious relationship we chose to use non-parametric regression to model the relationship. While there are several approaches for non-parametric regression, including K-nearest neighbors, alternating conditional expectations, and decision trees, our choice of model is dictated by the relatively small dataset. In situations such as these, a decision tree model might be a better option.

Image-Based Shape Attributes: Shape-based features are used to extract information related to the aspect ratio of the pores (ratio of the smallest to the largest axis) in the study. It is well known that low aspect ratio or lenticular pores are highly deformable in the direction perpendicular to the longest axis of the pores. The shape-based features are derived from the contours and boundaries of the identified pore segments in an image. While there are several shape attributes such as area, aspect ratio, circularity, solidity, extent, eccentricity, Euler number and orientation, we only use the aspect ratio of the pores. It is important to

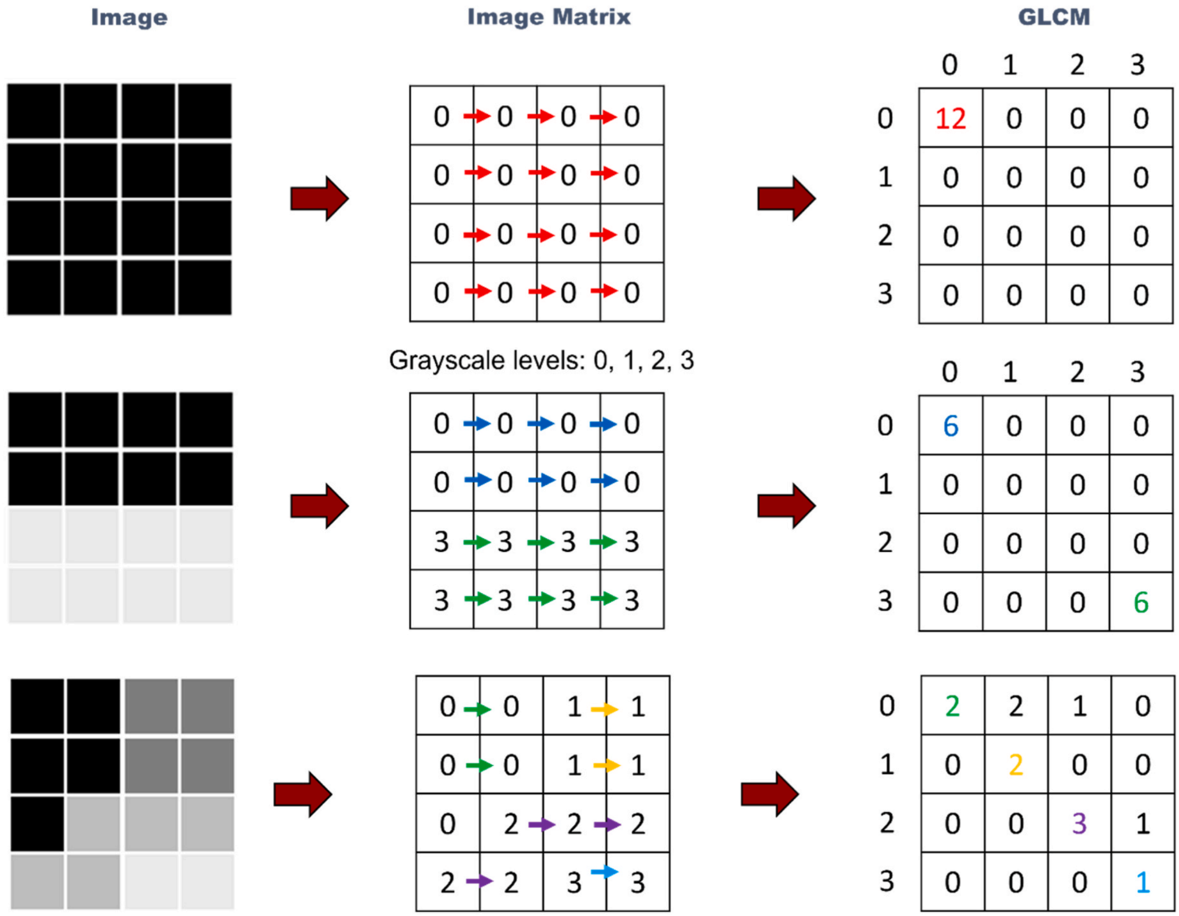

Fig. 3. The three rows on images show examples of the computation of the GLCM matrix. For simplicity, this example only considers neighboring pixels in the horizontal direction. When the image is uniformly black, the GLCM shows 12 pixels of intensity 0 adjacent to pixels of intensity 0. For the middle row, there are 6 pixels of intensity 0 adjacent to pixels of intensity 0 and another 6 pixels of intensity 3 adjacent to pixels of intensity 3. For the image at the bottom with heterogeneity, the GLCM matrix provides the corresponding spatial arrangement of grayscale values. Similar matrices can be constructed by considering neighboring pixels in other directions.

note that while most samples are characterized by pores with similar aspect ratios, there are samples where a broad range of pore aspect ratios exist. Fig. 5 shows the pore aspect ratio across two selected plays considered in this study. These two were chosen because in Fig. 5a associated with the Barnett shale, we mostly observe a similar set of aspect ratios while in Fig. 5b associated with the Woodford, we observe a broader distribution of pore aspect ratios and a significant percentage of low aspect ratio pores. It is likely that the stress response of a rock will be driven by the prevalence of the low aspect ratio pores.

Deep-Learning Features: Our study comprises of images from 14 different unconventional plays. In previous studies (Zhang et al., 2022; Knaup et al. 2023), we used a variant of AlexNet to classify images from various unconventional plays. The goal was to assess if deep learning algorithms can recognize and classify images based on their source. Deep learning networks for classification or any other computer vision task rely on successive sequences of convolution and pooling and various networks are associated with different architectures and model complexity. Although the network architectures might vary, the various trained models are associated with a key commonality. That is, the output derived following the last step of convolution and pooling is expected to contain features that are relevant to the classification problem. In the context of SEM image classification, these features are expected to contain information relevant to the unique microstructure associated with different images from different plays. Returning to Fig. 2, a human observer can visualize key differences in the microstructure of both images. In a trained deep learning-based classification network, the processed output will reflect the microstructural uniqueness embedded in an image.

In this study, we employed the ResNet-50 (He et al., 2016) model to classify the source of the images used in this study. The ResNet-50 model achieved 97 % accuracy and a very low misclassification rate in identifying the source of the SEM images on a balanced test dataset. The ResNet-50 model, when used as a feature extractor, produces a feature vector of length 2048 as seen in Fig. 6. The image in Fig. 6 shows four randomly selected feature maps from the 2048 7x7 feature maps that constitute the output of the last convolution layer. These features are often considered high-level and semantically meaningful, making them useful for various downstream tasks such as image classification, object detection, and image retrieval. The last global average pooling layer averages each of the feature maps to create a feature vector of length 2048 that contains the learned representations of the input image, prior to the final fully connected layer of the ResNet-50 architecture.

A typical image classification workflow involves standardization of the image grayscale to account for different acquisition settings, different instruments and even different instrument operators. We specifically standardized the grayscales used in the images prior to training the classifier. Other important aspects of constructing a trained classifier include the need for image augmentation to provide diversity of perspectives while training (Goodfellow et al., 2016). While there are several approaches for image augmentation, we created augmented views by flipping the images and rotating them to create related but different representations of the image.

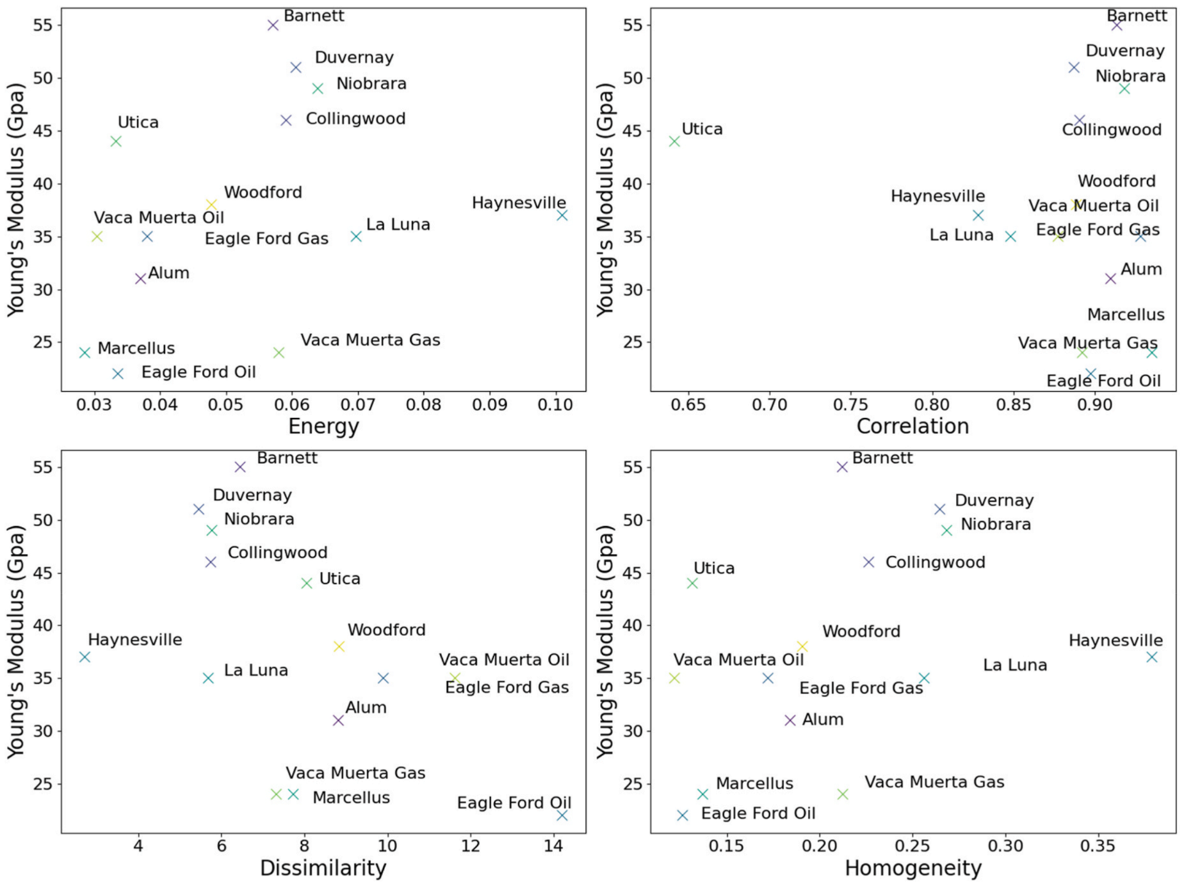

Fig. 4. Crossplots showing the measured Young's moduli versus the various textural attributes used in this study.

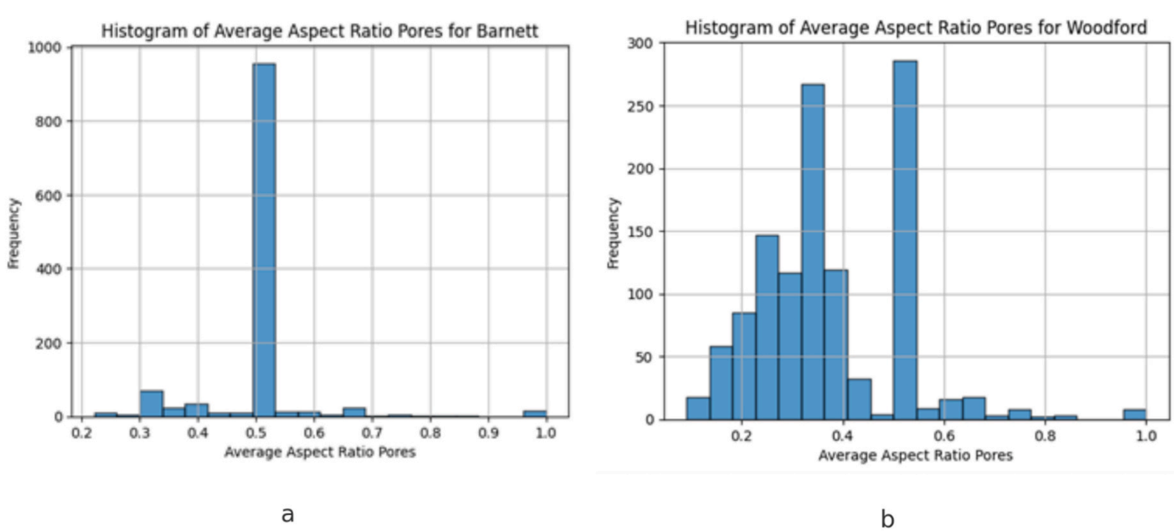

Fig. 5. The plot shows pore aspect ratios corresponding to the Barnett shale on the left in (a) and to the Woodford in (b) highlighting the presence of pores of different aspect ratios in different shales. Low aspect ratio pores are likely to be deformed more easily in the presence of an external stress, thereby impacting the mechanical behavior of the rock.

4. Model variants and description

We construct several models in order to assess how each attribute contributes to the estimation of Young's Moduli. The various attributes are textural attributes, pore shape attributes, and deep learning features from a trained classifier. All these models are trained on a decision tree regressor using LOOCV (Leave-One-Out-Cross-Validation). A decision tree regression model works by recursively splitting the data into subsets based on input feature values, which are the textural attributes and deep learning features in our study. The splits are made at values that maximize the homogeneity of the resulting leaf node. We made a choice of a decision tree because, firstly, our dataset necessitated a non-parametric regression model as seen in Fig. 4 and secondly, because of the smaller size of our dataset that limited the choice of non-parametric regression

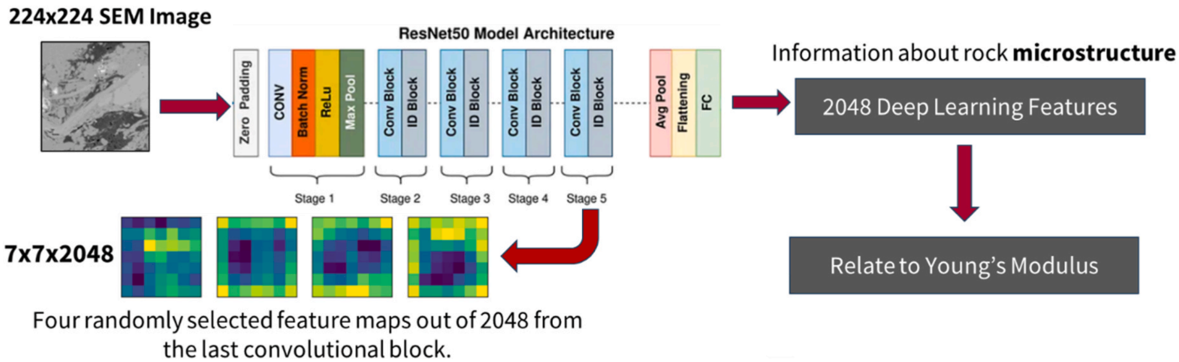

Fig. 6. The plot shows the entire workflow necessary for extracting the deep learning features. A 224x224 grayscale image passed through the ResNet50 network produces 2048 7x7 feature maps at the last convolutional layer. Subsequently, the global averaging pool averages each of the 2048 feature maps to form a vector of size 2048 which contains information related to the original image microstructure. These features are then related to the Young's Modulus as shown. We have implemented some form of dimensionality reduction to avoid the use of 2048 deep learning features. The choices we made include PCA (principal component analyses) and RP (random projections). The image of the ResNet50 is from https://upload.wikimedia.org/wikipedia/commons/9/98/ResNet50.png.

models.

The use of LOOCV is critical because the dataset comprises just 14 points. We also did not attempt more complex regression models because of the sparsity of the data. The models used in this study are as follows.

- Model-1: Uses only textural features.
- Model-2: Uses textural and shape features.
- Model-3: Uses textural, shape and deep learning features. Deep learning features are extracted from the trained ResNet-50 model. This model uses Principal Component Analysis for dimensionality reduction such that 99 % variance is preserved.
- Model-4: Uses textural, shape and deep learning features. Deep learning features are extracted from the trained ResNet-50 model. This model uses Random Projections (Dasgupta, 2000; Danala et al. 2022) for dimensionality reduction.

Principal components analysis (PCA) works by generating a linear combination of the input features that contain the highest variability in the dataset. In our work, the first 19 principal components contained over 99 % of the variability in our input data. The method of random projections (Dasgupta, 2000) works by identifying a lower dimensional space where the distance between the input datapoints is preserved. This occurs by using a random projection matrix between the higher and lower dimensions. If the dimensions are large enough, most random features are approximately orthogonal (Dasgupta, 2000) and so, the structure in the data is preserved.

5. Model validation

Model validation is a critical step in the machine learning workflow to assess the performance of a predictive model. We perform model validation using the Root Mean Squared Error (RMSE) as the performance metric and Leave-One-Out Cross-Validation (LOOCV) for model evaluation. LOOCV is suitable for smaller datasets to maximize the amount of training data and where neither test/train splits nor k-fold cross-validation can be performed.

Leave-One-Out Cross-Validation (LOOCV) is an exhaustive cross-validation technique that involves using all but one observation as training data and the last remaining observation/datapoint for validation. We then repeat the process of validation by with a choice of another datapoint for validation and the remaining datapoints for training. This is repeated such that each observation in the sample is used exactly once as the validation data. In the end, LOOCV requires 'N' such model fits where 'N' is the number of datapoints. In this manner, LOOCV verifies that the model is generalizable to new measurements within the range of the original data.

Together, RMSE and LOOCV create a comprehensive approach to validating machine learning models, ensuring that the models are not only accurate but also robust and generalizable across different data points within the dataset.

6. Results and analyses

Following LOOCV using a decision tree, we obtain the results shown in Table 2 which provides the RMSE values for prediction as well as the standard deviations in RMSE values during the LOOCV. It is also expected, for a reasonable predictive model, that the standard deviation be low with respect to the absolute value of the RMSE. A high standard deviation in the predictions during k-fold cross validation or leave-one-out cross validation is indicative of a model that cannot generalize very well to new data.

Model 2, which uses both textural features as well as accounting for pore shape, outperformed all other models. Not only was the RMSE value the lowest, but across the LOOCV, the standard deviation in the RMSE was low. This is important to mention because a more robust model not only possesses a higher prediction accuracy but is also relatively insensitive to the choice of test data. This is true for both LOOCV and during k-fold cross validation.

To illustrate why the variability in the results needs to be considered, we can compare Models 3 and 4 which use the same input features but rely on PCA or Random Projections for the dimensionality reduction. The use of random projections (Model 4) seems to outperform its PCA counterparts (Model 3) when considering the lowest RMSE value. However, the variability in the RMSE is also relatively higher implying that the model does not generalize well to new (or test) data, which is a prerequisite for selecting the best model. In principle, random

Table 2
The list of all four models used in this study and the corresponding RMSE. The standard deviation represents variations in RMSE with LOOCV. It is expected that this value be low related to the absolute value of the RMSE. PCA stands for Principal Components Analyses and RP is an abbreviation for Random Projections.

Model	Best RMSE	Mean RMSE	Standard Deviation
Model 1 (Texture Features)	8.48	9.99	0.56
Model 2 (Texture and Shape-based Features)	6.75	8.63	0.70
Model 3 (Texture and Shape-based and Deep Learning Features with PCA)	11.77	12.39	0.42
Model 4 (Texture and Shape-based and Deep Learning Features with RP)	12.40	14.77	1.19

projections are an effective alternative to PCA when dealing with high dimensionality of the data or computational constraints. However, in this study, random projections can be considered to be the least effective approach for dimensionality reduction.

The importance of maintaining low variability across validation is also highlighted in Fig. 7. The figure plots the RMSE values across LOOCV with the different models used in the study. The lowest RMSE value obtained across all models during LOOCV is associated with the model using random projections paired with texture, shape, and deep learning attributes. However, we can also see that the variations during LOOCV are significantly higher. For this study, therefore, we can see that using both textural and shape-based features outperform all other models. The use of texture alone is generally satisfactory, but the addition of shape-based features enhances the accuracies. An important thing to note is that the addition of any deep-learning features from a trained classifier did not aid the prediction of Young's modulus in any way.

Fig. 8 shows the model predictions against the measured Young's modulus for the various plays demonstrating the applicability of our texture-based approach for unconventional plays.

7. Generalizability of the approach

While we demonstrate the suitability of a texture-based approach to predict Young's moduli from images in unconventional shales, we also recognize that the approach comes with a few limitations. First, this approach is likely not to be applicable to samples that are relatively homogeneous. The samples in this study are typical of unconventional shales that are highly heterogeneous and fine-grained. Secondly, it is important that very tight quality controls be imposed on the acquisition of the nanoindentation measurements that form our target variable. If the nanoindenter only probes a single larger grain or material, then the measured elastic properties will not be representative of the entire sample. It is crucial that the elastic moduli data be representative of the sample by taking multiple measurements across a grid of points and averaging the data.

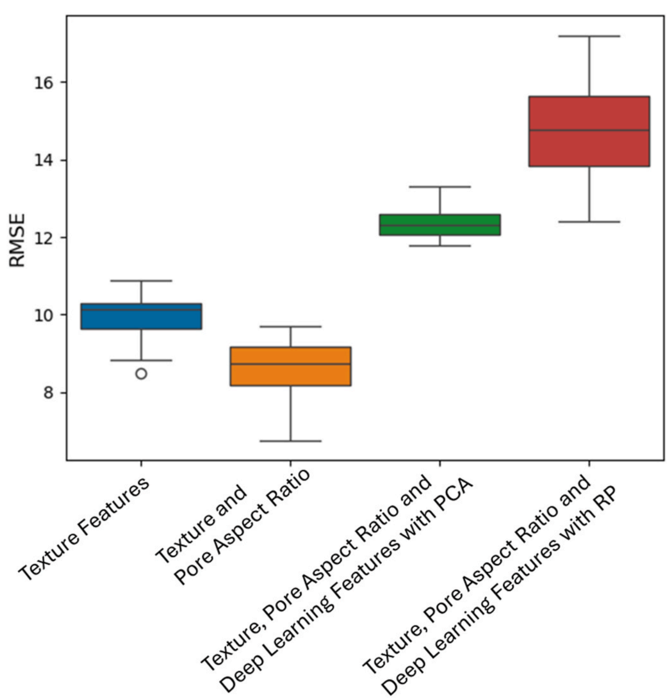

Fig. 7. A plot comparing the performance of the models across different feature groups. The use of texture and shape-related features results in the lowest average RMSE as well as the smallest variability during LOOCV. This supports our assertion that the use of texture is important for the prediction of mechanical properties and that it is important to account for the shapes of the pores as well. In the figure, PCA stands for Principal Component Analysis and RP for Random Projections.

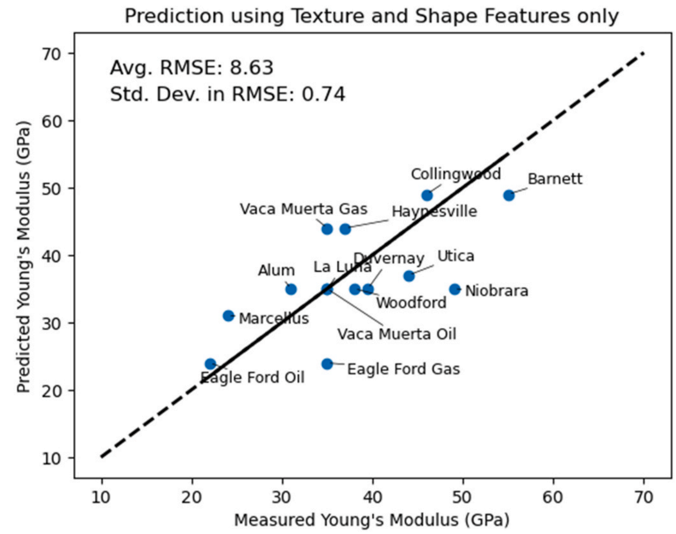

Fig. 8. A plot of the predicted Young's modulus versus the actual Young's modulus using the best model obtained in this study. The best model uses textural attributes and pore aspect ratios as inputs for prediction. The colors used for the data are merely to distinguish the various points. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

On the other hand, this approach can easily be generalized to other applications such as mining, tunneling or any other activity where a knowledge of mechanical properties is essential, but the acquisition of those properties is time-consuming and/or inconvenient, especially in remote locations. As we mentioned earlier, the basic idea can be utilized to obtain elastic properties for more coarse-grained rocks using thin-section images. Additionally, while the current application focuses on mechanical properties, the approach can potentially be extended to the prediction of other petrophysical properties such as porosity and mineralogy.

8. Conclusions

This work demonstrates a promising approach to predict rock elastic properties for unconventional shales. The paper describes an approach using texture as well as pore aspect ratios acquired from BSE-SEM images to relate to the measured Young's moduli. The following conclusions can be drawn from this study.

1. The use of pore aspect ratios allows us to capture the effect of lenticular pore systems that impact the elastic properties of rocks.
2. Texture is shown to be sufficient to capture the mineralogy as well as the juxtaposition of various minerals in the fabric of the rock.
3. Deep learning features from a network trained to recognize the microstructure and source of the images do not add any additional information to the prediction of elastic properties. This reinforces our initial hypothesis that the elastic properties are functions of the arrangement of the minerals that can be captured by texture alone.

CRediT authorship contribution statement

Sai Kiran Maryada: Visualization, Software. **Deepak Devegowda:** Writing – original draft, Supervision, Investigation, Conceptualization. **Chandra Rai:** Writing – review & editing, Investigation, Conceptualization. **Mark Curtis:** Data curation. **David Ebert:** Writing – review &

editing, Validation. **Gopichandh Danala:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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